BEYOND MULTIPLE LINEAR REGRESSION · WEEK 2

When treatment effects vary

Interactions, random slopes and crossed random effects

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METHODS · DESIGN · STATISTICS

Assignment 1: comparing protein extraction protocols

THE EXPERIMENT

- Question: does a modified protocol increase protein yield?
- Tissue from each of 10 donors is divided into eight aliquots.
- Within each donor, four aliquots are randomly assigned to each protocol: Standard or Modified.
- Each aliquot is measured once, giving 80 observations.
- Outcome: protein yield in micrograms per milligram of tissue (µg/mg).

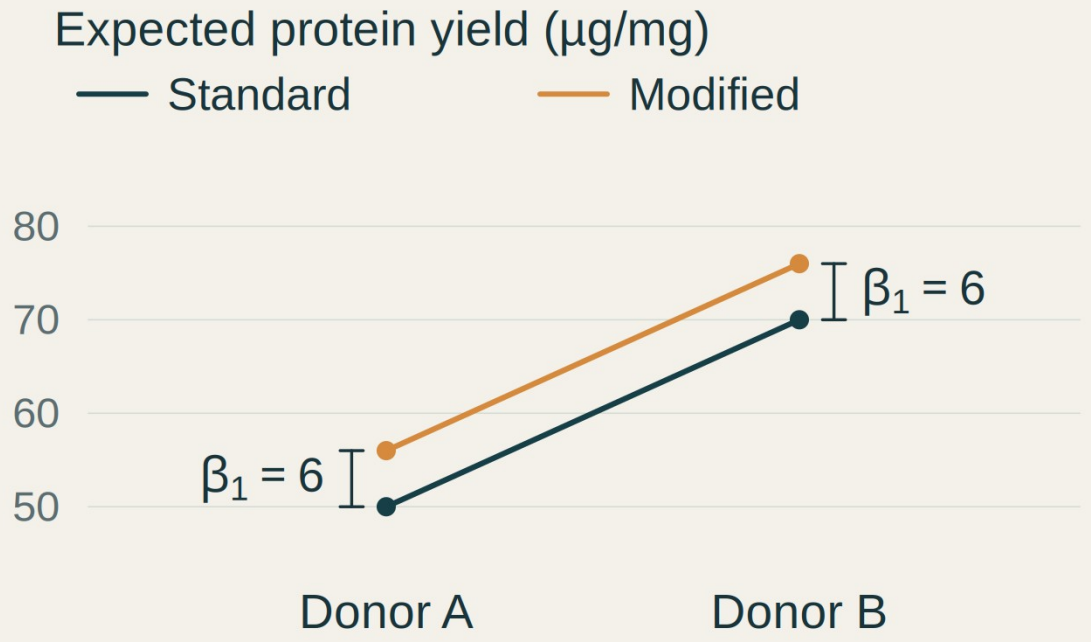
Randomized block design
Block: donor; experimental unit: aliquot.



The additive model: a common protocol effect across donors

Illustration with two donors and a true protocol effect of 6 µg/mg in each donor.

Donor	Standard	Modified	Row average
A	$\mu_{0A} = 50$	$\mu_{1A} = 56$	**53**
B	$\mu_{0B} = 70$	$\mu_{1B} = 76$	**73**
Column average	$\beta_0 = 60$	66	**63**



μ_{tj}: true mean for protocol t in donor j

$$Y_{ij} = \underbrace{\beta_0 + c_j}_{\text{Standard-protocol mean for donor } j} + \underbrace{\beta_1 T_{ij}}_{\text{Protocol contribution}} + \underbrace{\varepsilon_{ij}}_{\text{Aliquot deviation}}$$

T_{ij} = 0 for Standard and 1 for Modified. i = aliquot; j = donor.

Individual aliquot yields vary around their group mean through ε_{ij}.

Assignment 1 recap: comparing the models

Modified-minus-standard effects, estimated SEs and 95% CIs in µg/mg.

Model	Estimate	Estimated SE	95% CI	p-value
Protocol only	4.61	2.89	-1.15 to 10.38	0.115
Fixed donor effects	4.61	1.76	1.10 to 8.12	0.011
Random donor effects	4.61	1.76	1.10 to 8.12	0.011

All three models estimate the same protocol effect. Accounting for donor reduces the estimated standard error.

Assignment 1: feedback on your submissions

PLACEHOLDER

To be completed after reviewing the submissions.

Two or three observations about the analyses, interpretations and reporting.

Changing protocol allocation within donors

ASSIGNMENT 1 ALLOCATION

Donor	Standard	Modified
1	4	4
2	4	4
3	4	4
4	4	4
5	4	4
6	4	4
7	4	4
8	4	4
9	4	4
10	4	4
Total	40	40

HYPOTHETICAL ALLOCATION

Donor	Standard	Modified
1	1	7
2	2	6
3	3	5
4	3	5
5	4	4
6	4	4
7	5	3
8	5	3
9	6	2
10	7	1
Total	40	40

Only allocation changes: the true protocol effect remains 6 µg/mg in every donor.

The models now give different protocol estimates

Unequal allocation within donors; estimated modified-minus-standard difference in protein yield.

Model	Estimate (µg/mg)	Estimated SE (µg/mg)
Protocol only	5.65	2.90
Fixed donor effects	4.44	1.95
Random donor effects	4.53	1.94

Accounting for donor changes the estimate. The fixed and random donor models give slightly different estimates, with similar estimated standard errors.

The models use different comparisons

PROTOCOL ONLY

- Compare Standard and Modified across all aliquots.
- The groups may contain different mixtures of donors, so their difference also reflects donor baselines.

FIXED DONOR EFFECTS

- Compare Standard and Modified within each donor.
- Aliquots in each comparison share the same donor baseline.

RANDOM DONOR EFFECTS

- Within donors: compare Standard and Modified in the same donor.
- Between donors: relate average yield to the proportion assigned to Modified.

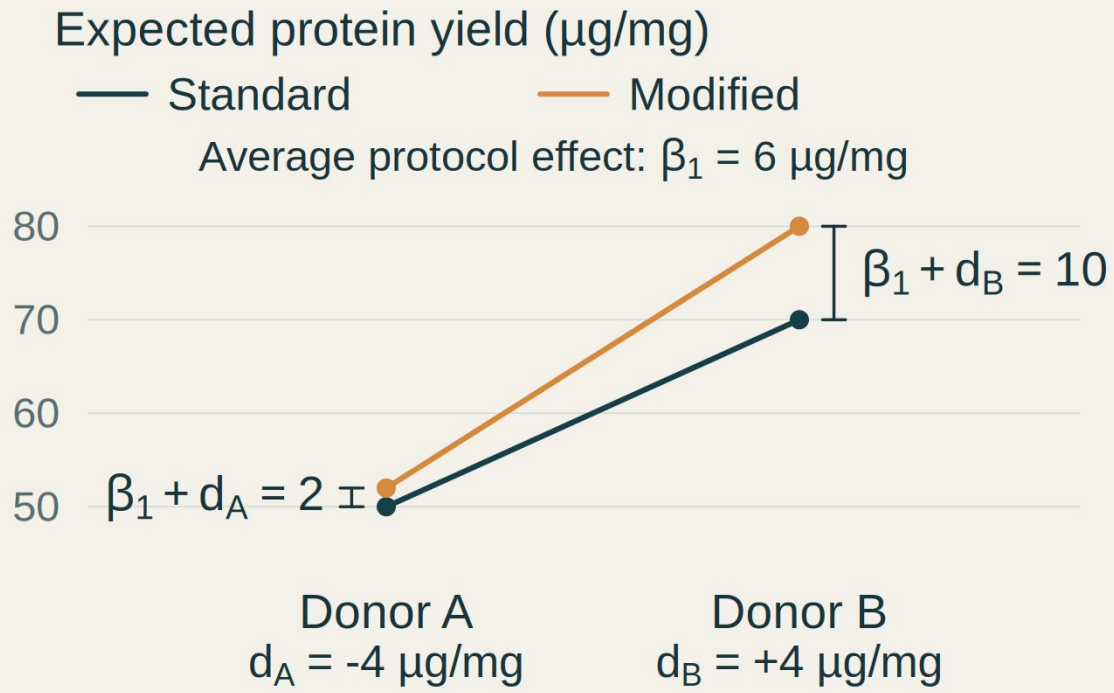
Both sources contribute to the protocol-effect estimate.

So far: the same protocol effect in every donor. What if protocol effects vary?

Different protocol effects across donors

Illustration with two donors and true protocol effects of 2 and 10 µg/mg.

Donor	Standard	Modified	Row average
A	$\mu_{0A} = 50$	$\mu_{1A} = 52$	**51**
B	$\mu_{0B} = 70$	$\mu_{1B} = 80$	**75**
Column average	$\beta_0 = 60$	66	**63**



μ_{tj}: true mean for protocol t in donor j

$$Y_{ij} = \underbrace{\beta_0 + c_j}_{\text{Standard-protocol mean for donor } j} + \underbrace{(\beta_1 + d_j)T_{ij}}_{\text{Protocol contribution}} + \underbrace{\varepsilon_{ij}}_{\text{Aliquot deviation}}$$

T_{ij} = 0 for Standard and 1 for Modified. i = aliquot; j = donor.

Individual aliquot yields vary around their group mean through ε_{ij}.

How should we represent differences between groups?

Donors, hospitals and other groups can differ in baseline outcome and treatment effect.

FIXED GROUP EFFECTS

Give each included group its own baseline deviation and treatment-effect deviation.

Express the average treatment effect across the included groups.

RANDOM GROUP EFFECTS

Describe baseline deviations and treatment-effect deviations through a population distribution.

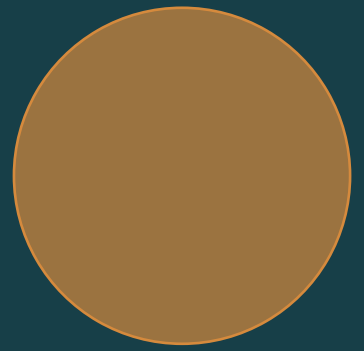
Express the average treatment effect in the population represented by those groups.

We start with fixed effects for groups, using a multicenter clinical trial.

PART 1

Treatment-by-center interaction: fixed effects





Case study: CPAP and blood pressure

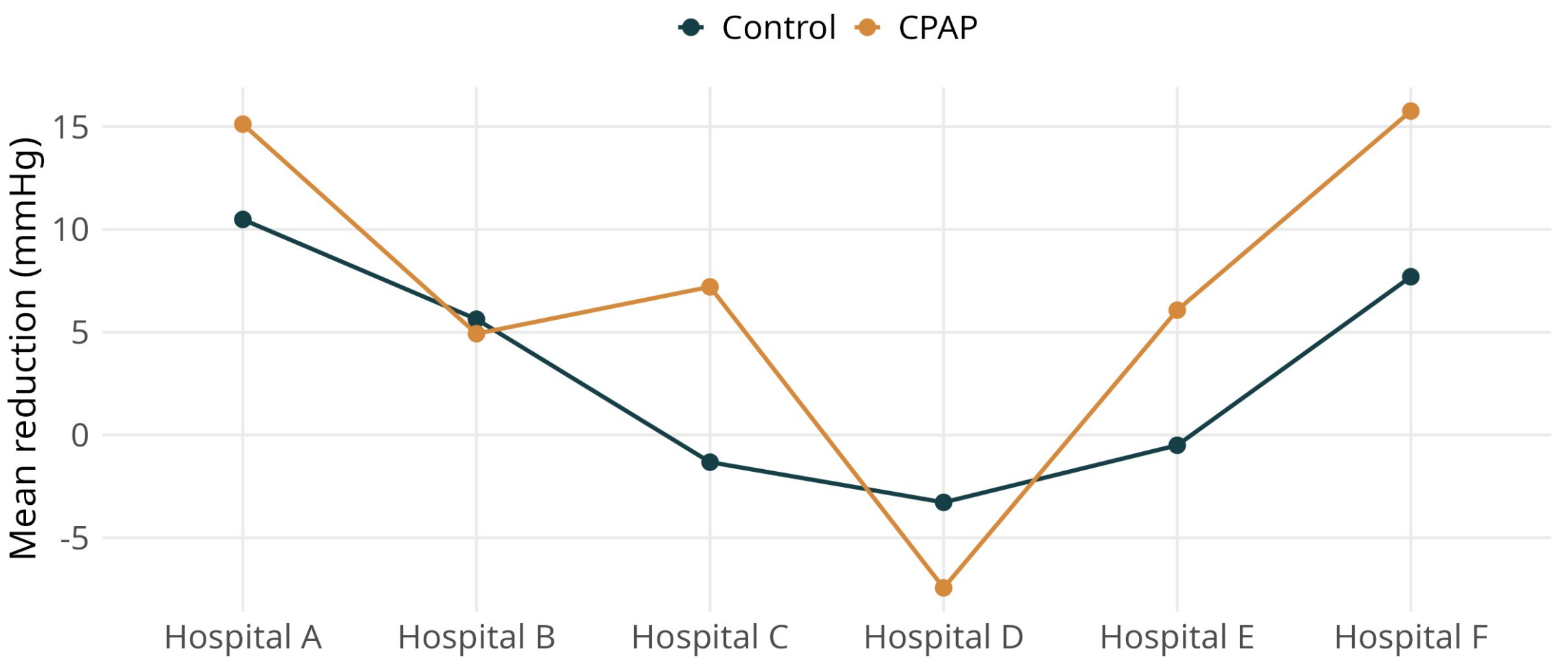
A SIMULATED MULTICENTER TRIAL

- 240 patients with sleep apnea and hypertension despite medication, across six hospitals.
- Hospital sizes: 24, 32, 36, 44, 48 and 56; equal treatment allocation within each.
- Control: continue usual blood-pressure medication.
- CPAP: add continuous positive airway pressure through a mask during sleep.
- Outcome: reduction in 24-hour mean blood pressure after 12 weeks (mmHg).

Positive values indicate a blood-pressure reduction. Inspired by HIPARCO (Martínez-García et al., JAMA, 2013); simulated patient data.



Blood-pressure reduction by treatment and hospital



A fixed-effects model for treatment and center

$$Y_{ij} = \beta_0 + c_j + (\beta_1 + d_j)T_{ij} + \varepsilon_{ij}$$

Y_{ij}: blood-pressure reduction (mmHg). T_{ij} = 0 for control and 1 for CPAP.

CONTROL OUTCOMES

β_0: average control-group reduction across the included centers.

c_j: center j's deviation from that control average.

$$\sum_j c_j = 0$$

TREATMENT EFFECTS

β_1: average treatment effect across the included centers.

d_j: center j's deviation from the average treatment effect.

$$\sum_j d_j = 0$$

Each center receives equal weight in the averages. Center deviations are fixed, unknown parameters.

Fixed-effects interaction model: results

Term	Estimate (mmHg)	Estimated SE (mmHg)	95% CI (mmHg)	p-value
Intercept (β_0)	3.1	0.75	1.6 to 4.6	< .001
CPAP (β_1)	3.8	1.06	1.7 to 5.9	< .001
Hospital A	7.4	2.00	3.4 to 11.3	< .001
Hospital B	2.5	1.77	-1.0 to 6.0	0.157
Hospital C	-4.4	1.69	-7.8 to -1.1	0.009
Hospital D	-6.4	1.56	-9.5 to -3.3	< .001
Hospital E	-3.6	1.51	-6.6 to -0.6	0.017
CPAP × Hospital A	0.8	2.82	-4.8 to 6.4	0.774
CPAP × Hospital B	-4.5	2.50	-9.5 to 0.4	0.072
CPAP × Hospital C	4.7	2.39	0.0 to 9.4	0.050
CPAP × Hospital D	-8.0	2.20	-12.3 to -3.6	< .001
CPAP × Hospital E	2.8	2.13	-1.5 to 7.0	0.198

Effects coding: Hospital F follows from the sum-to-zero constraints.

Estimated marginal means

Estimated marginal means (EMMs) are model-based means, averaged over selected factors.

1. DEFINE THE TREATMENT–HOSPITAL GRID

Treatment	A	B	C	D	E	F		Row average
Control	$\widehat{\mu}_{0A}$	$\widehat{\mu}_{0B}$	$\widehat{\mu}_{0C}$	$\widehat{\mu}_{0D}$	$\widehat{\mu}_{0E}$	$\widehat{\mu}_{0F}$	→	$\widehat{\mu}_{0\cdot}$
CPAP	$\widehat{\mu}_{1A}$	$\widehat{\mu}_{1B}$	$\widehat{\mu}_{1C}$	$\widehat{\mu}_{1D}$	$\widehat{\mu}_{1E}$	$\widehat{\mu}_{1F}$	→	$\widehat{\mu}_{1\cdot}$

2. PREDICT EACH CELL MEAN

Use the fitted model for each treatment–hospital combination.

$$\widehat{\mu}_{tj} = \widehat{\beta}_0 + \widehat{c}_j + (\widehat{\beta}_1 + \widehat{d}_j)t$$

t = 0 for Control; t = 1 for CPAP.
j identifies the hospital.

3. AVERAGE ACROSS HOSPITALS

Average each row, giving all six hospitals equal weight.

$$\widehat{\mu}_{t\cdot} = \frac{\widehat{\mu}_{tA} + \widehat{\mu}_{tB} + \cdots + \widehat{\mu}_{tF}}{6}$$

This gives one estimated marginal mean per treatment.

Which treatment effect do we want to estimate?

MARGINAL TREATMENT EFFECT

How do treatments compare on average across hospitals?

Compare the two treatment means averaged equally across the six hospitals.

With our effects-coded model, the estimated marginal treatment effect is $\widehat{\beta}_1$.

SIMPLE TREATMENT EFFECTS

How do treatments compare within a particular hospital?

Compare the estimated CPAP and control means within that hospital.

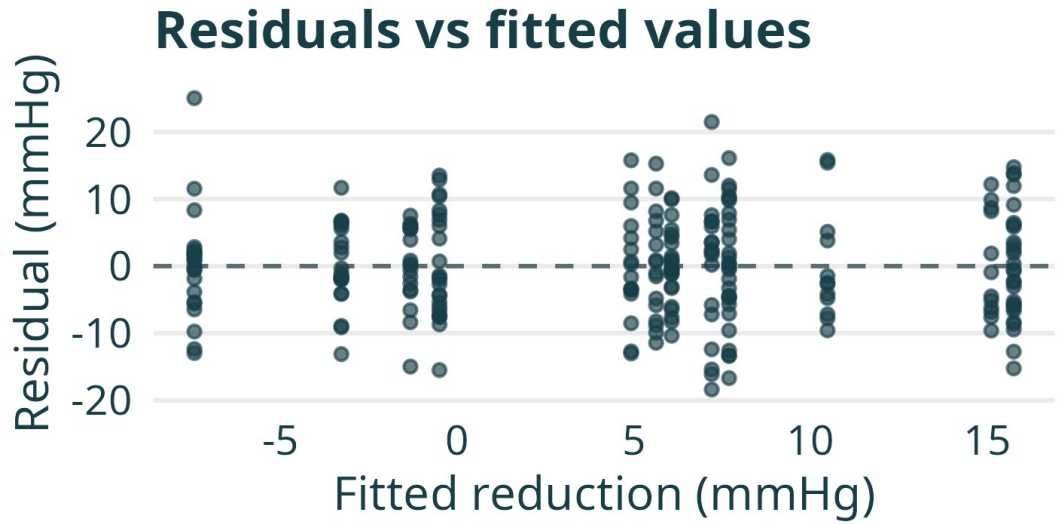
With our effects-coded model, the estimated simple treatment effect in hospital j is $\widehat{\beta}_1 + \widehat{d}_j$.

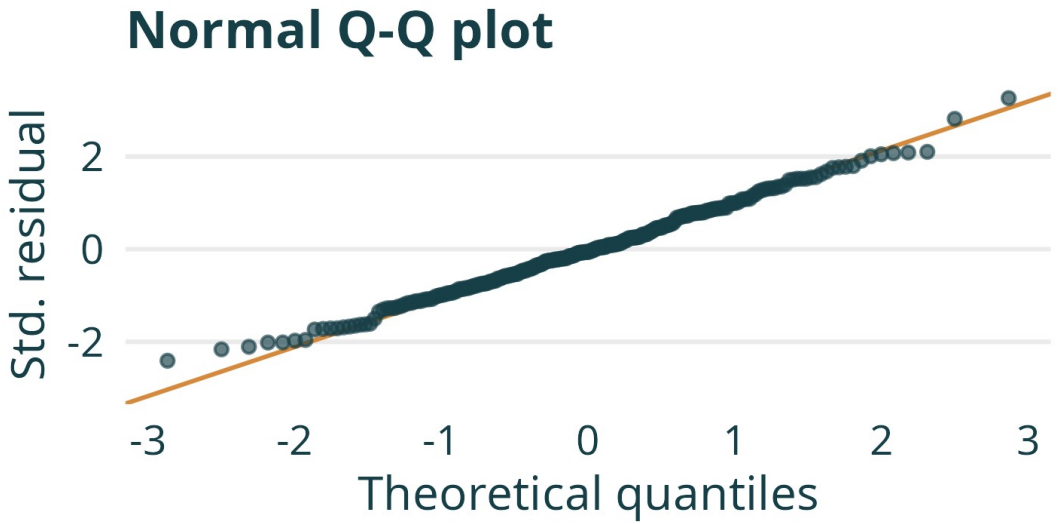
Estimated means and simple treatment effects

	Estimated means		Simple treatment effect		
Hospital	Control	CPAP	Difference	95% CI	p-value
A	10.5	15.1	4.6	-1.7 to 11.0	0.150
B	5.6	4.9	-0.7	-6.2 to 4.8	0.800
C	-1.3	7.2	8.5	3.4 to 13.7	0.001
D	-3.3	-7.4	-4.2	-8.8 to 0.5	0.081
E	-0.5	6.1	6.6	2.1 to 11.0	0.004
F	7.7	15.8	8.1	3.9 to 12.2	< .001

Means, differences and CIs in mmHg. Difference: CPAP minus control.
Pointwise 95% CIs and unadjusted p-values.

Model diagnostics: fixed-effects interaction





WHAT WE USE THIS PLOT FOR

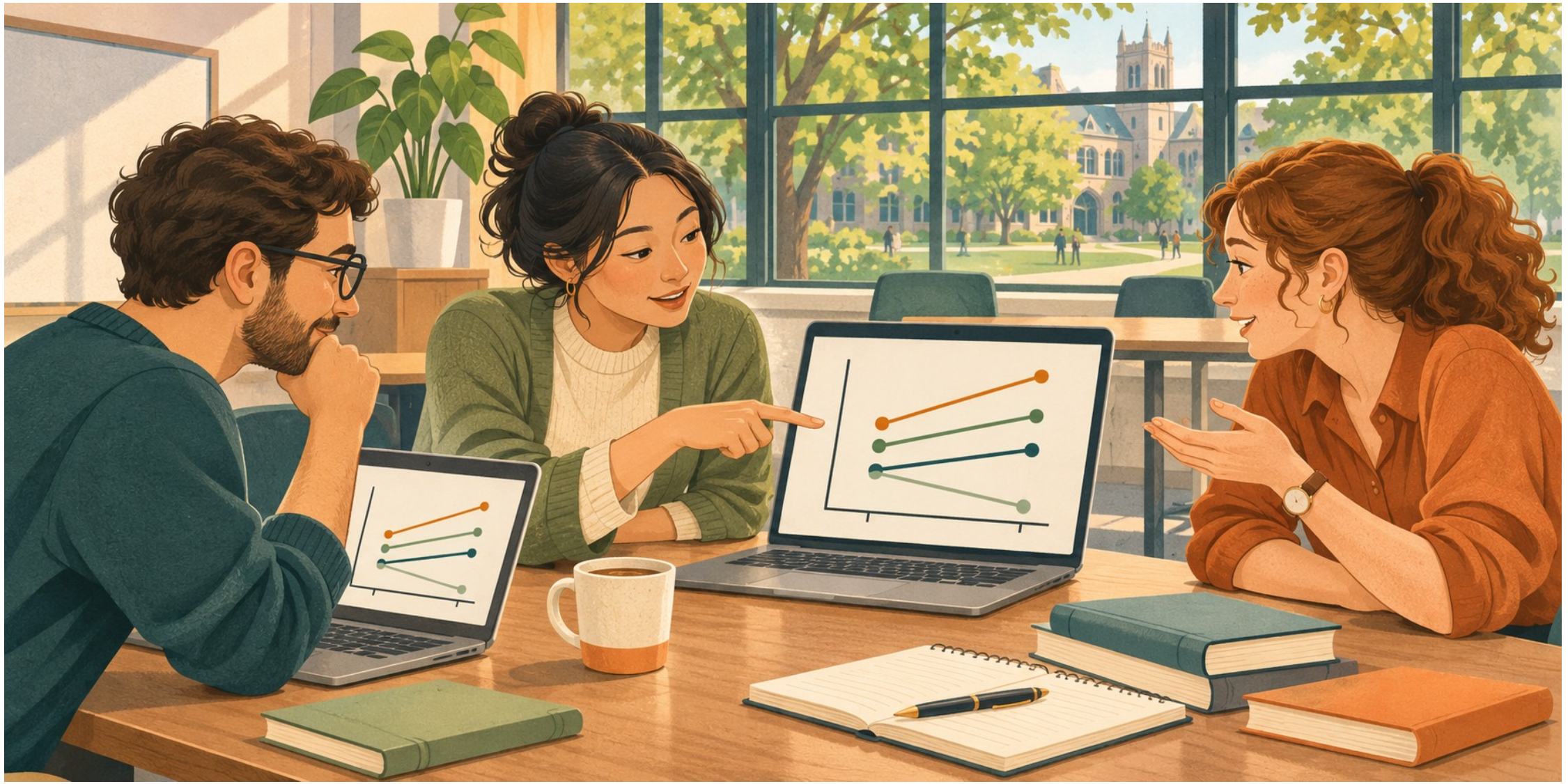
Assess whether residual spread is similar across fitted blood-pressure reductions.

WHAT WE USE THIS PLOT FOR

Assess whether the residual distribution is approximately Normal.

Residual spread is broadly similar across fitted values, and points lie close to the Normal reference line.

Time for Lab 3



Lab 3: what if we leave out the interaction?

Average effect across the five laboratories
Experimental minus standard, in percentage points of wound closure

Model	Estimate	Estimated SE	95% CI	p-value
Additive	9.28	1.81	5.69 to 12.88	< .001
With interaction	9.28	1.62	6.07 to 12.49	< .001

Same estimate: 10 plates per treatment in every laboratory make both models estimate the average of the five within-laboratory treatment differences.

Larger estimated SE: without interaction, differences in treatment effects remain in the residuals, increasing the estimated residual variation.

In this balanced study, omitting interaction leaves the average treatment estimate unchanged but increases its estimated uncertainty.

What if laboratory sizes differ?

WHAT THE MODELS ESTIMATE

- With our effects coding, the interaction model estimates the equally weighted average treatment effect, regardless of laboratory sizes.
- If laboratories have equal sizes, the additive model estimates this same average.
- If laboratory sizes differ, the additive model gives larger laboratories more weight.

HYPOTHETICAL EXAMPLE

Effects in percentage points

Laboratory	Plates	Treatment effect
Small	20	5
Large	80	15

Interaction model: 10 points

Additive model: 13 points

When treatment effects differ across laboratories, omitting interaction can change both the interpretation of the treatment coefficient and its estimated uncertainty.

PART 2

Treatment-by-center interaction: random effects

From the included centers to a population

THESE SIX HOSPITALS

Imagine repeating the study with new patients in the same six hospitals.

The centers' true treatment effects stay the same.

We estimate the average effect in these six hospitals.

A POPULATION OF CENTERS

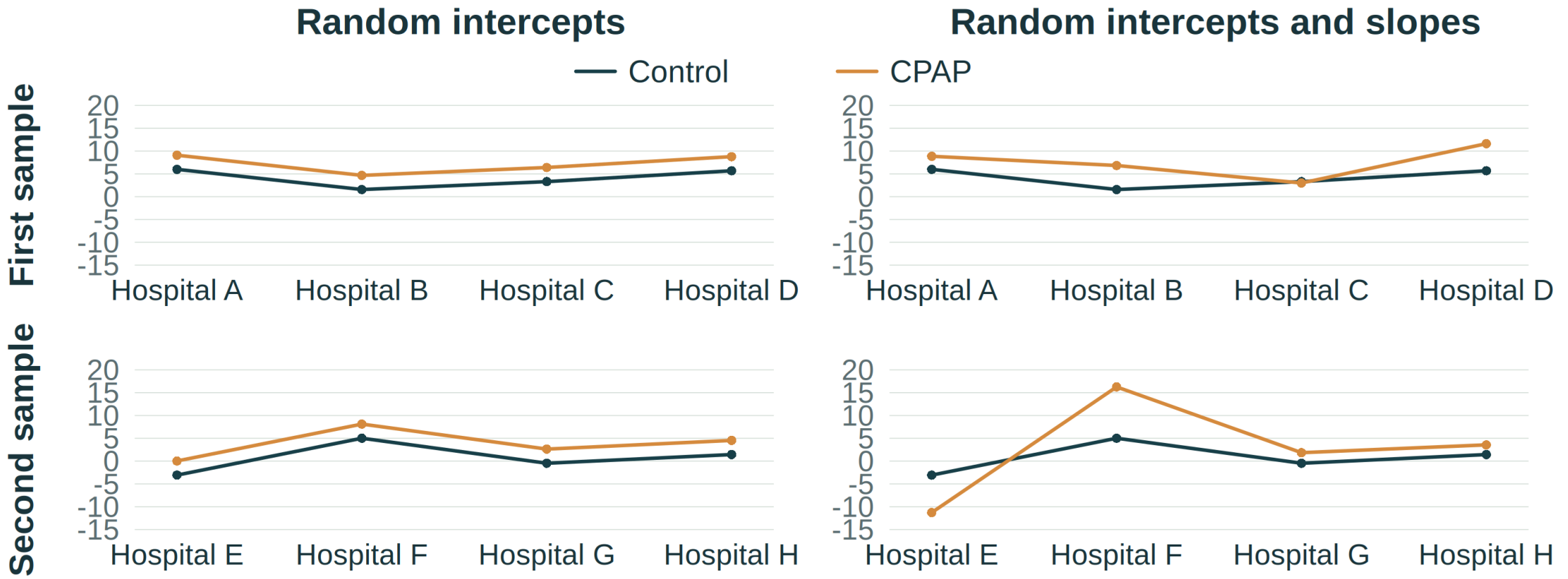
Imagine repeating the study with new centers from the same population.

The selected centers can have different true treatment effects.

We estimate the average effect across the population of centers.

When we generalize to a population of centers, uncertainty also depends on which centers enter the study.

What changes when we sample new centers?



Hypothetical true mean blood-pressure reductions (mmHg) across hospitals.

The population model stays the same, but each sample contains centers with different realized effects.

The center effects follow a population distribution

$$Y_{ij} = \beta_0 + c_j + (\beta_1 + d_j)T_{ij} + \varepsilon_{ij}$$

β_0 and β_1 are population averages. T_{ij} = 0 (control) or 1 (CPAP).

DISTRIBUTIONAL ASSUMPTIONS

$$\begin{pmatrix} c_j \\ d_j \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_c^2 & \sigma_{cd} \\ \sigma_{cd} & \sigma_d^2 \end{pmatrix}\right]$$

$$\varepsilon_{ij} \sim N(0, \sigma^2)$$

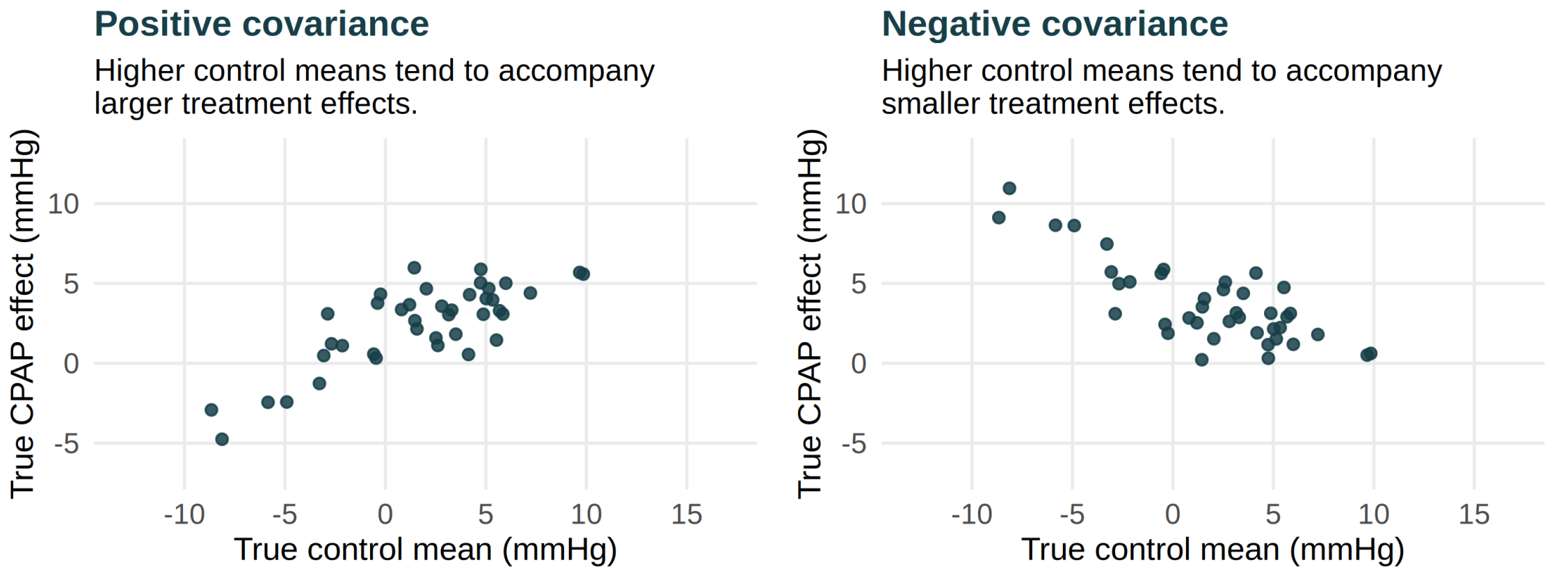
Effect pairs are independent across centers. Errors are mutually independent and independent of the center effects.

VARIANCES AND COVARIANCE

σ_c^2: variance of control means across hospitals.
σ_d^2: variance of treatment effects across centers.
σ_{cd}: covariance of control-mean and treatment-effect deviations.
σ^2: individual variance around the center-specific treatment mean.

Center deviations have population means of zero. Their average varies across samples of hospitals.

How do control outcomes and treatment effects vary together?



True control means and CPAP-minus-control effects (mmHg). Each point is one hypothetical hospital.

Covariance describes how a hospital's control outcome and treatment effect vary together.

CPAP study: estimated variance components

Results from the random-intercept and random-slope model fitted to the simulated CPAP data.

Component	Parameter	Estimate (mmHg²)
Control means across hospitals	σ_c^2	26.2
Treatment effects across centers	σ_d^2	20.5
Control-mean / treatment-effect covariance	σ_{cd}	11.2
Individual residual variation	σ^2	61.8

Estimated control-mean / treatment-effect correlation

$$\widehat{\rho}_{cd} = \frac{\widehat{\sigma}_{cd}}{\sqrt{\widehat{\sigma}_c^2 \widehat{\sigma}_d^2}} = \frac{11.2}{\sqrt{26.2 \times 20.5}} \approx 0.48$$

Hospitals with greater control-group reductions tend to have larger treatment effects.

Treatment-effect variance is comparable to control-mean variance. Both are smaller than individual residual variance.

The average treatment effect across the population of centers

The same 240 patients in six hospitals, fitted with random intercepts and treatment slopes.

Population parameter	Estimate	Estimated SE	95% CI	p-value
Average control-group reduction (β_0)	2.99	2.22	-2.7 to 8.7	0.237
Average treatment effect (β_1)	3.99	2.12	-1.4 to 9.4	0.117

Estimates, estimated SEs and confidence intervals are in mmHg.

The treatment coefficient estimates the average CPAP-minus-control effect across the population of centers.

Estimated average benefit: 4.0 mmHg. The 95% confidence interval includes zero.

Fixed and random effects: estimates and uncertainty

Hospital sizes differ, with equal treatment allocation within every hospital.

Model	Estimated treatment effect (mmHg)	Estimated SE (mmHg)	p-value
Fixed interaction	3.82	1.06	< .001
Random intercept and slope	3.99	2.12	0.117

Fixed interaction: equally weighted average for these six hospitals. Uncertainty comes from sampling patients.

Random intercept and slope: population-average effect. Uncertainty also reflects which hospitals entered the study.

Unequal hospital sizes allow different estimates. Between-hospital treatment variation adds uncertainty about the population average.

What if we omit the random treatment slope?

Back to the CPAP study: both models target the population-average treatment effect.

Model	Estimated treatment effect (mmHg)	Estimated SE (mmHg)	p-value
Random intercept only	4.08	1.05	< .001
Random intercept and slope	3.99	2.12	0.117

Random intercept only: assumes the same treatment effect in every center.

Random intercept and slope: allows treatment effects to vary across centers.

When treatment effects vary across centers, a random-intercept-only model can underestimate uncertainty about the population-average effect.

Time for Lab 4



Variance decomposition depends on the random-effects structure

WEEK 1: RANDOM INTERCEPT

The same center contribution applies under both treatments.

Outcome variance separates into a center component and an individual component.

This gives one common within-center correlation: the ICC.

WEEK 2: RANDOM INTERCEPT AND SLOPE

The center contribution depends on treatment.

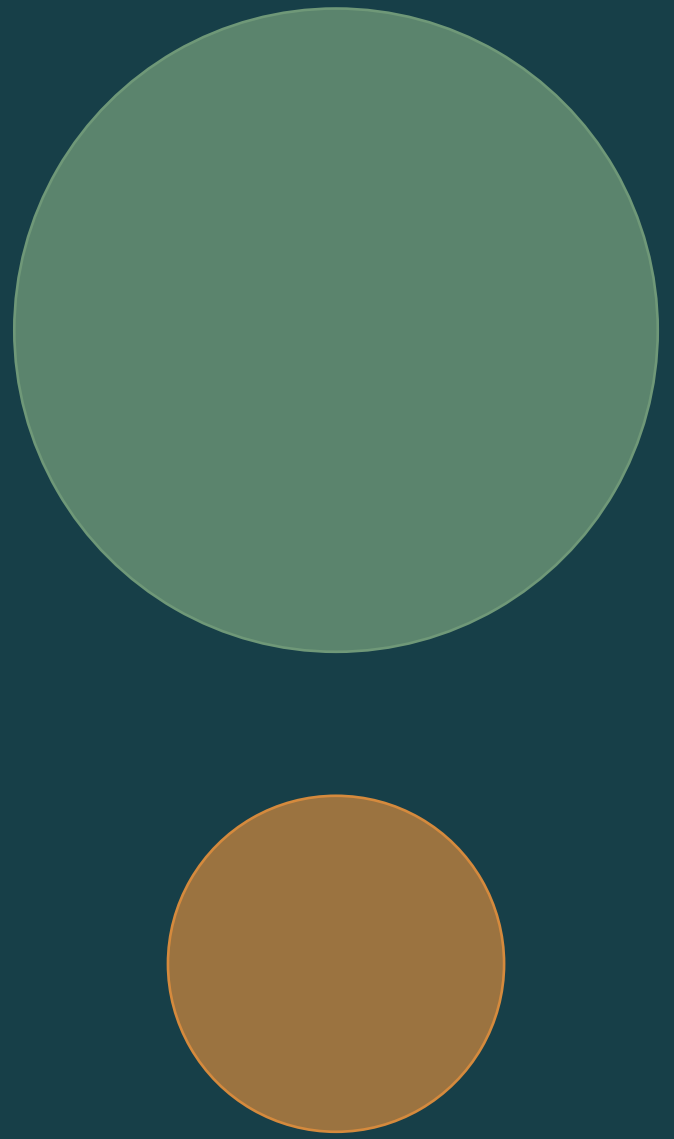
Outcome variance depends on treatment and combines the relevant variances and covariance.

Within-center correlation depends on the treatments received by the two patients.

Variance decomposition still applies, but there is no longer one common variance breakdown or a single ICC for all patients.

PART 3

Crossed random effects



What if there is more than one grouping variable?

So far: one grouping variable, center. What changes with two or more grouping variables?

NESTED GROUPING VARIABLES

Each group at one level belongs to exactly one group at the next level.

Patients belong to wards, and each ward belongs to one hospital.

Grouping variables: ward and hospital.

CROSSED GROUPING VARIABLES

Neither grouping variable is contained within the other.

Each patient is assessed by several radiologists, and each radiologist assesses several patients.

Grouping variables: patient and radiologist.

Next: crossed grouping variables, illustrated by patients and radiologists.
Week 3: nested grouping variables, such as wards within hospitals.

Case study: agreement between radiologists

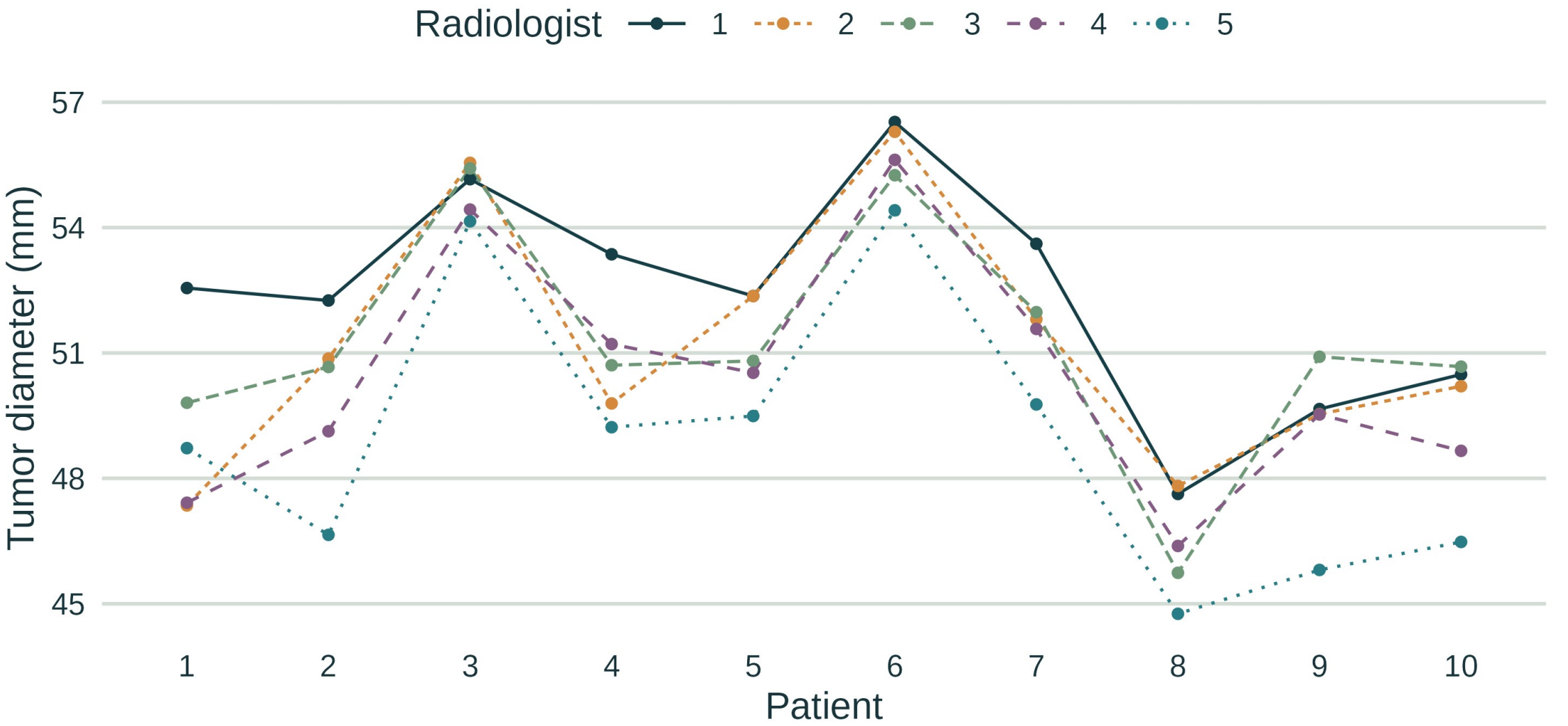
MEASURING TUMOR DIAMETER

- Simulated study: CT scans from 10 patients assessed independently by five radiologists.
- Every radiologist measures the maximum tumor diameter on every scan.
- Outcome: tumor diameter in millimeters, giving 50 measurements.
- We want to generalize beyond these patients and radiologists.

Do different radiologists obtain similar measurements for the same patients?



Exploring the radiologists' measurements



A model with two crossed grouping variables

$$Y_{ij} = \mu + u_i + v_j + \varepsilon_{ij}$$

Y_{ij} is the measurement for patient i made by radiologist j.

WHAT CONTRIBUTES TO A MEASUREMENT?

μ: overall mean tumor diameter.

u_i: patient i's deviation from the mean.

v_j: radiologist j's tendency to measure higher or lower.

ε_{ij}: remaining measurement variation.

TWO RANDOM INTERCEPTS

$$u_i \sim N(0, \sigma^2_{\text{patient}})$$

$$v_j \sim N(0, \sigma^2_{\text{radiologist}})$$

$$\varepsilon_{ij} \sim N(0, \sigma^2)$$

Patient effects, radiologist effects and residual errors are mutually independent.

Measurements are related when they share the same patient or the same radiologist.

Where does measurement variation come from?

Estimated variance components from the crossed model fitted to the radiologist study.

Source of variation	Estimate (mm^2)	Interpretation
Patient	7.44	Differences between patients
Radiologist	1.47	Systematic radiologist differences
Residual	1.00	Remaining measurement variation

$$\widehat{\mathrm{ICC}} = \frac{7.44}{7.44 + 1.47 + 1.00} \approx 0.75$$

Intraclass correlation coefficient (ICC): absolute agreement for a single measurement.

About 75% of the estimated total variation is between patients. Radiologist differences and residual variation reduce agreement.

The End

Questions



